

Asynchronous Programming & Concurrency for AI

Parallelism → At same time run multiple ops

Multi-processing → CPU cores (parallelly)

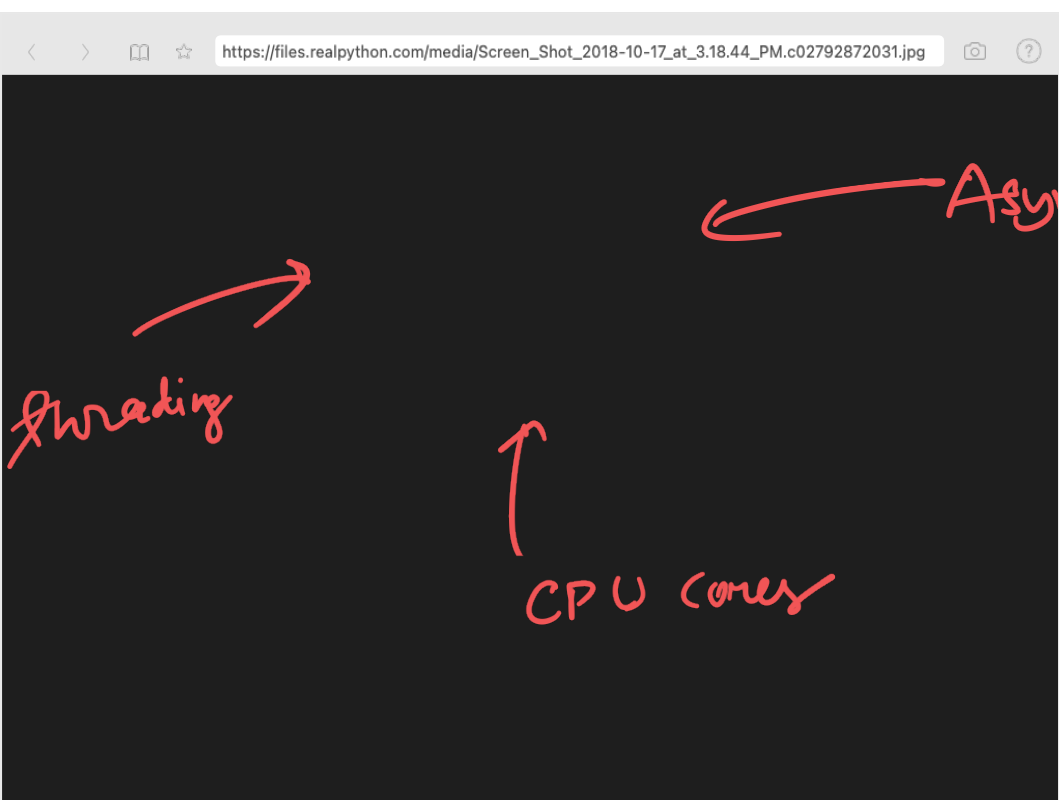
Concurrency → Broader term -

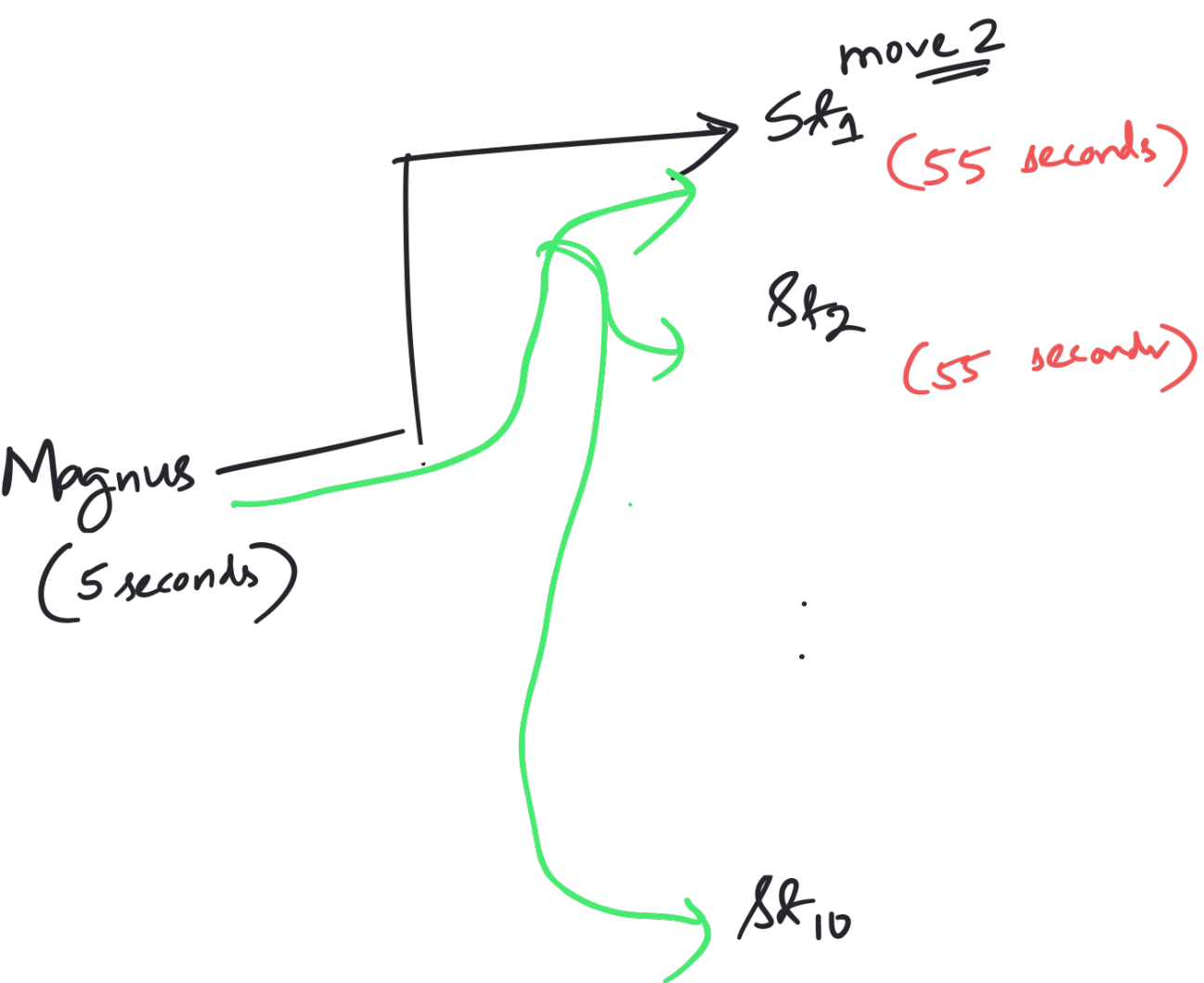
Threading → Use multiple threads

Type of tasks :

I/O Bound (I/O to compute)

CPU Bound (Utilizing CPU cons to max)





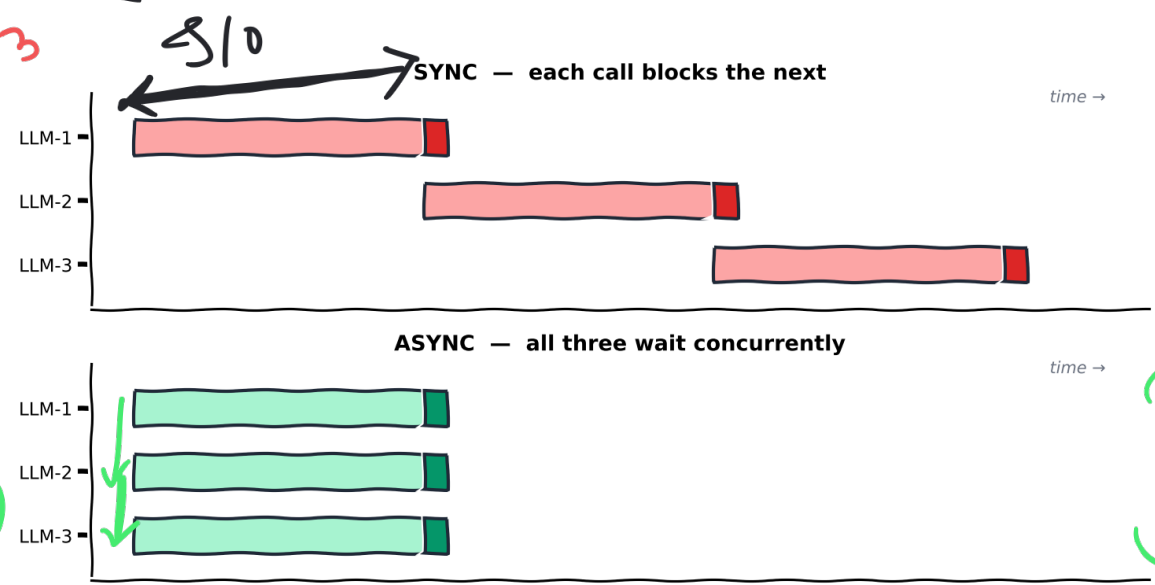
30 moves
 $30 * (55 + 5) = 1800$

$5 * 10 = 50$

time = $t_1 + t_2 + t_3$

time = $\max(t_1, t_2, t_3)$

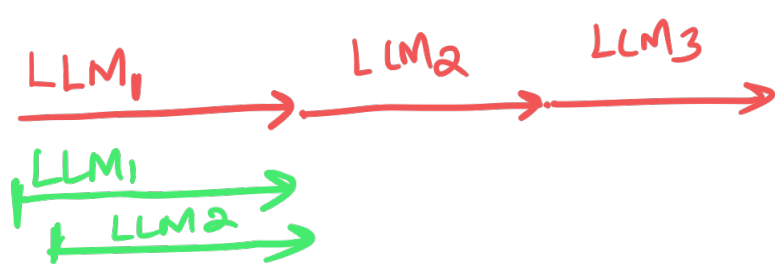
Open AG server



3 calls - $\left\{ \begin{array}{l} t_1 \in LLM_1 \\ t_2 \in LLM_2 \\ t_3 \in LLM_3 \end{array} \right\}$

No async programming

async



Non - Blocking

non-blocking

await

asyncio.sleep(0.1)

- pause the coroutine (0.1) so
- during that time the event loop can run other tasks

another process can execute

sleep(0.1)

← Blocking

async sleep

import sleep

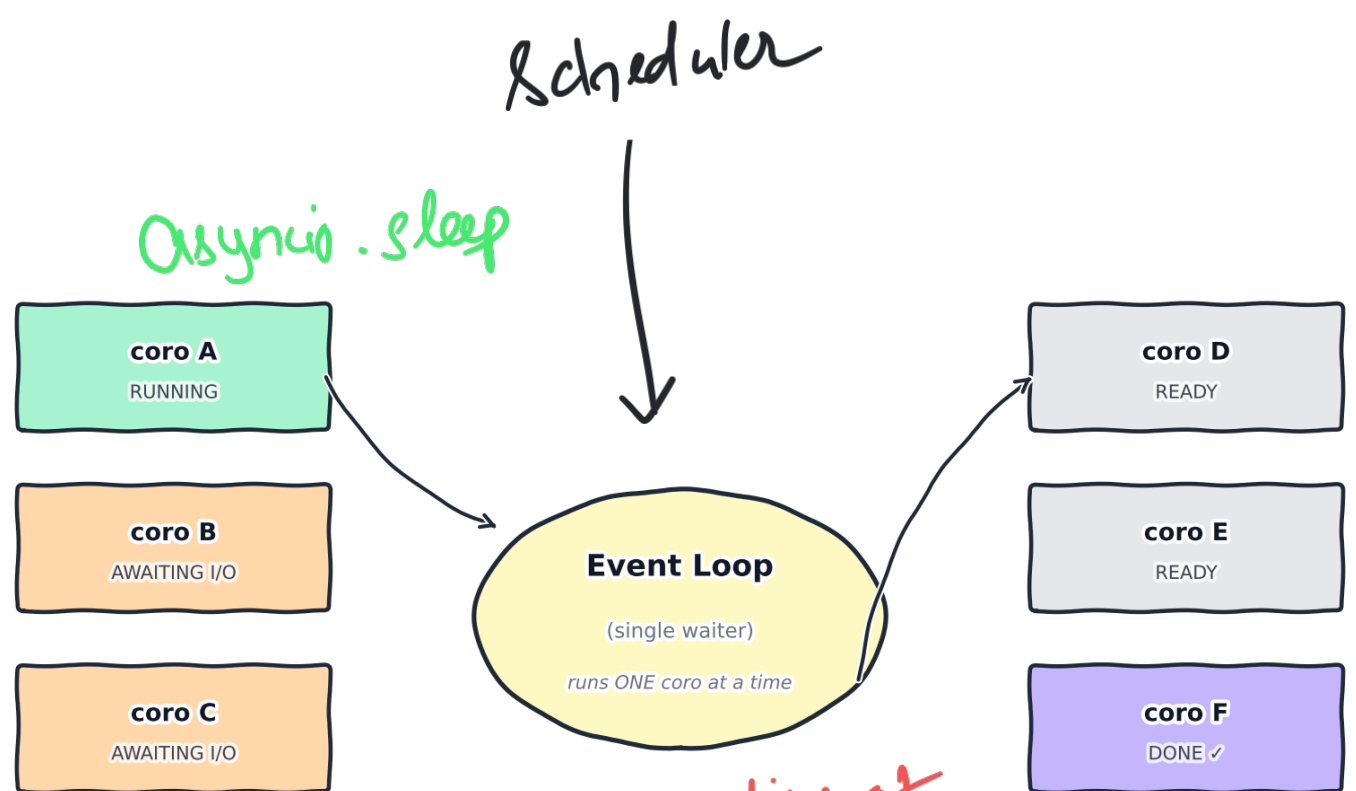
print(A)

sleep(1s)

print(B)

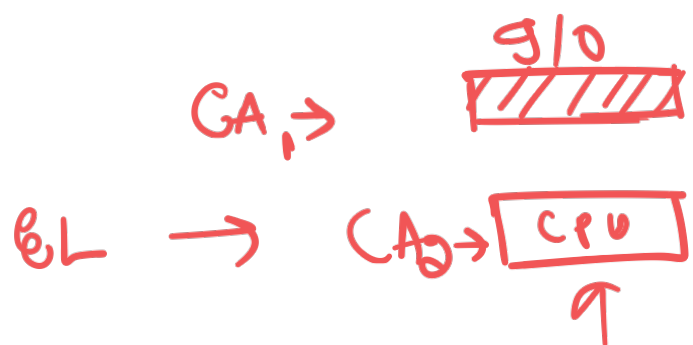
Block

Goroutine fun
goroutine obj
Event loop



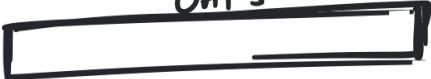
run one coroutine at a time

on await → park it • I/O done → ready • loop picks next ready coro



CA₁ → LLM_{OAI} ("Prompt 1")

✓ CA₂ → LLM_{OAI} ("Prompt 2")

CA₁ →  S/O (Waiting for OAI server response)



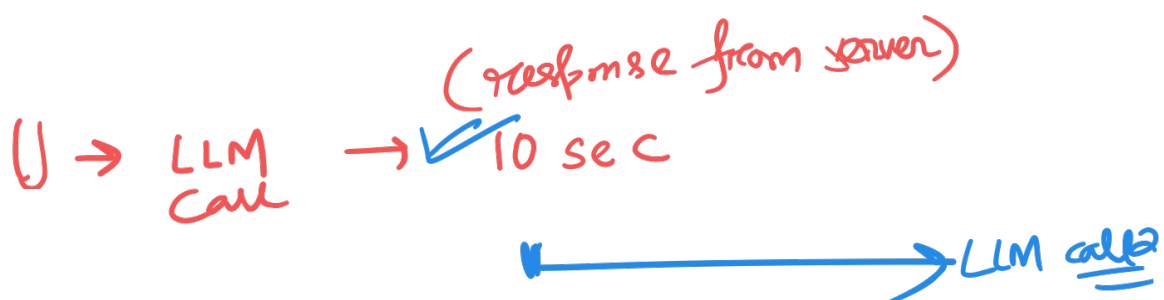
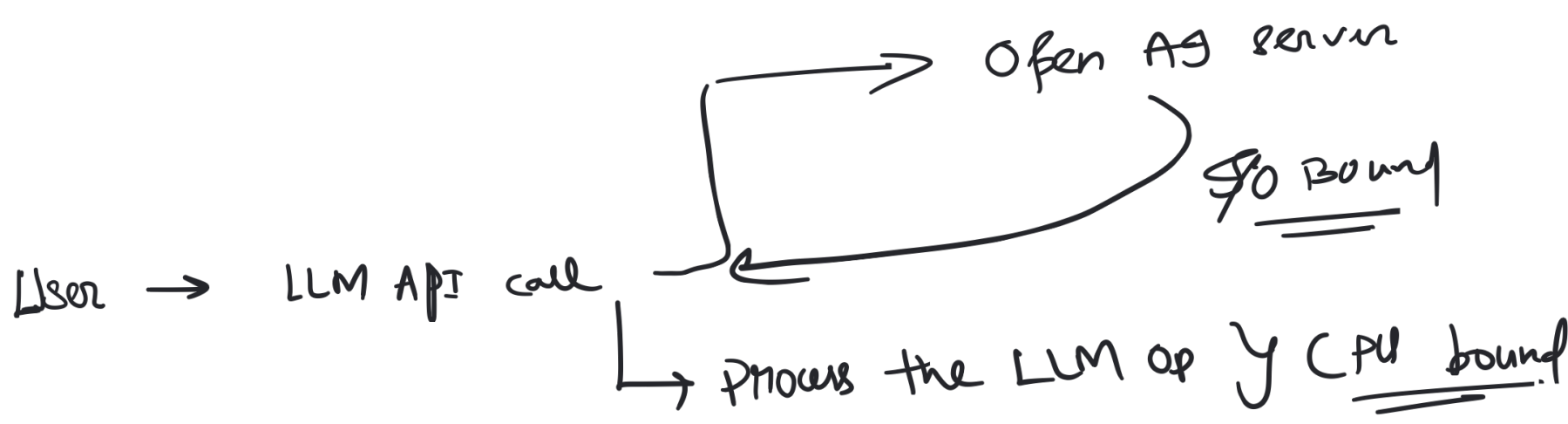
CA₂ → 

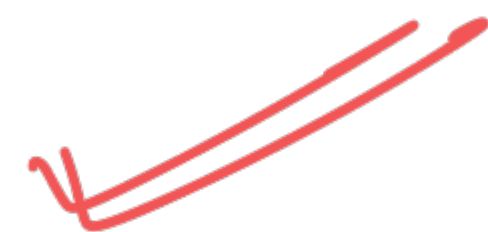
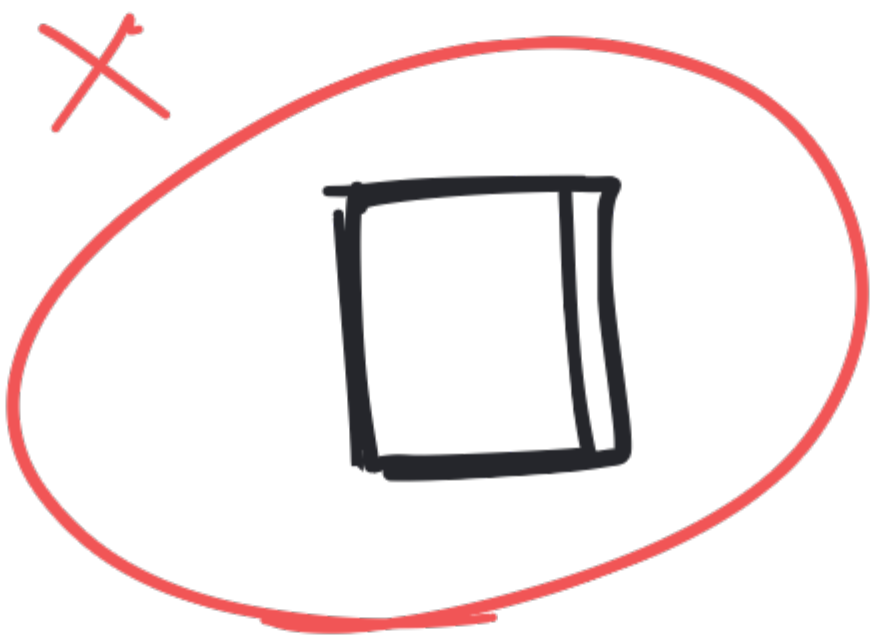
} Scheduler
(Event loop)

Asyncio. gather - runs multiple awaitable objects
Schedules for concurrent execution on event loop
results are in same order
all results at once / specific order



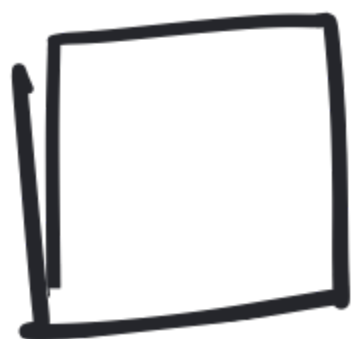
9:15



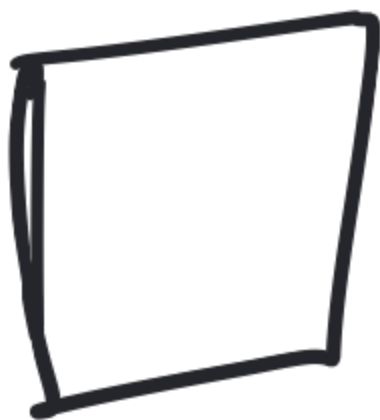


A₁

Elem failure

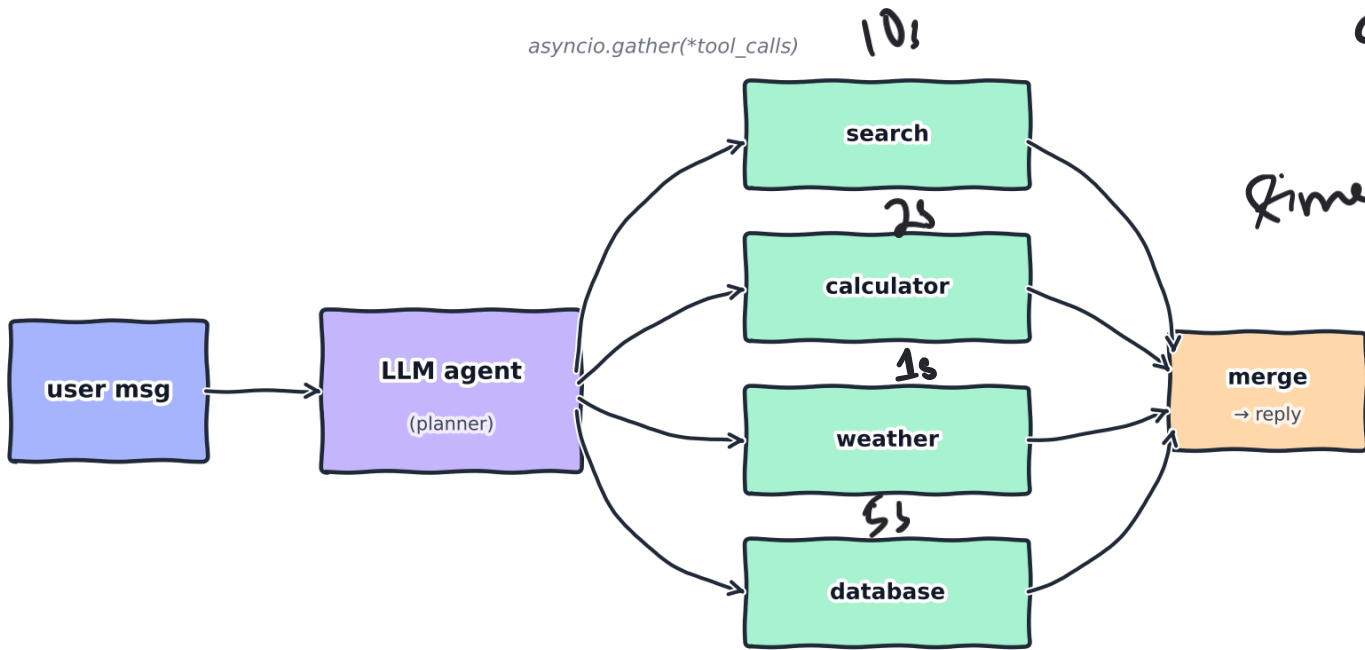


A₂



A₃

Search - web
calc
weather
db



Time = 10s
search, calc, w, db

$$T = 10 + 2 + 1 + 5$$

S → C → W → D

Concurrently

S
C
W
D

✓ Inference →

✓ Audio ~~AD~~ → later

Langchain
Langgraph
✓ deepagents

} Claude code

→ 2 classes

Pydantic

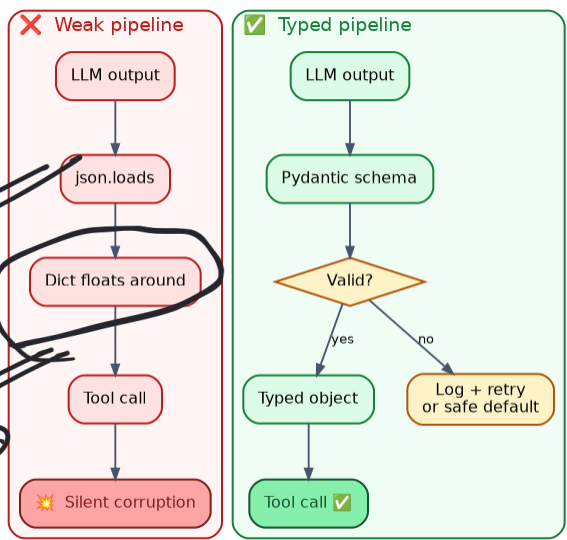


LLM op → adhere to some schema

json.loads

{ "phi": 10 }

{ "phi": 10 }



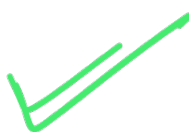
"Weather API ("Tokyo")

~~WeatherAPI(10)~~ ✗

LLM outputs

deterministic

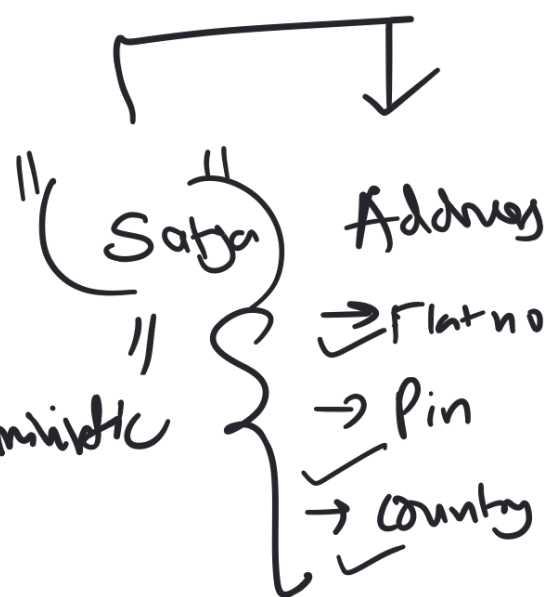
varying



User expecting OP for an app of struct



has to be deterministic



Country →